Data Structures

Part 2: Tuples

Too-pulls? Tuh-pulls? Too-plét?

Regardless, storing groups of data together that cannot be changed.

Tuples

A tuple is a data structure that stores multiple values in a specific order, similar to a list.

Unlike a list, a tuple is immutable, meaning its values cannot be added, removed, or changed after the tuple is created.

A tuple is created by separating values with commas (`,`), usually inside parentheses (`()`).

Tuples are useful for storing collections of values that should remain unchanged.

```
colors = ("red", "green", "blue")


print(colors)
```

OUTPUT

```
('red', 'green', 'blue')
```

Similarities Between Tuples and Lists

Tuples and lists are both ordered collections, meaning their values are stored in a specific order.

Like lists, tuple values can be accessed using positive or negative indices and sections can be accessed using slicing.

You can traverse a tuple with a **`for`** loop, check membership using **`in`**, and find its length using **`len()`**.

The main difference is that a tuple cannot be modified after it is created.

```
colors = ("red", "green", "blue")

for color in colors:
    print(color)

if "green" in colors:
    print("Green is in the tuple!")

print(len(colors))
```

Mutability vs Immutability

A mutable object can be changed after it is created, while an immutable object cannot.

Lists are mutable, so their items can be added, removed, or replaced.

Tuples are immutable, so their items cannot.

If a tuple contains a mutable object (like a list), the mutable object within the tuple can still be modified as normal.

```
data = ("Alice", [0, 1, 2])

# Lists are mutable
data[1].append(3)

print(data)
```

OUTPUT

```
('Alice', [0, 1, 2, 3])
```

Mutability vs Immutability *continued...*

Trying to change a tuple directly results in an error.

```
data = ("Alice", [0, 1, 2])

# Not allowed
data.append(100)

print(data)
```

```
OUTPUT
AttributeError: 'tuple' object has no attribute 'append'
```